

# Compact-Domain AMLEs with Arbitrary Complete $\mathbb{R}$ -Tree Targets

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## Status

This packet records a candidate partial result, likely valid, toward the conjectural extension stated after Theorem 1 of Naor and Sheffield [1]. The source proves existence and uniqueness of tree-valued absolutely minimal Lipschitz extensions when the domain is locally compact and the target tree comes from a finite graph. It conjectures that local compactness can be removed and that the target may instead be a bounded  $\mathbb{R}$ -tree. We remove the finiteness restriction on the target for compact domains, and prove existence. We do not prove uniqueness for the enlarged target class or remove local compactness of the domain.

## Source Passage

$$\mathrm{Lip}_U(h) \geq \mathrm{Lip}_U(\tilde{f}). \quad (1)$$

In other words,  $\tilde{f}$  extends  $f$ , and it is not possible to modify  $\tilde{f}$  on an open set in a way that decreases the Lipschitz constant on that set.

Our main result is:

**Theorem 1.** *Let  $X$  be a locally compact length space and let  $T$  be a metric tree. For every closed subset  $Y \subseteq X$ , every Lipschitz mapping  $f : Y \rightarrow T$  has a unique AMLE  $\tilde{f} : X \rightarrow T$ .*

Recall that a metric space  $(X, d_X)$  is a **length space** if for all  $x, y \in X$ , the distance  $d_X(x, y)$  is the infimum of the lengths of curves in  $X$  that connect  $x$  to  $y$ . By a **metric tree** we mean the one-dimensional simplicial complex associated to a finite graph-theoretical tree with arbitrary edge lengths (i.e., a finite graph-theoretical tree whose edges are present as actual intervals of arbitrary length, equipped with the graphical shortest path metric). We did not investigate here the greatest possible generality in which Theorem [1](#) holds true; in particular, we conjecture that the assumption that  $X$  is locally compact can be dropped, and that  $T$  need not correspond to a finite graph-theoretical tree, but rather can belong to the more general class of bounded  $\mathbb{R}$ -trees (see [\[14\]](#) [\[15\]](#)). The requirement that  $X$  be locally compact is not used in our proof of the uniqueness assertion of Theorem [1](#).

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## Question and Answered Part

The source defines a metric tree as the one-dimensional simplicial complex of a finite graph-theoretical tree, with arbitrary positive edge lengths. It then writes:

We conjecture that the assumption that  $X$  is locally compact can be dropped, and that  $T$  need not correspond to a finite graph-theoretical tree, but rather can belong to the more general class of bounded  $\mathbb{R}$ -trees.

We establish the target-tree extension when  $X$  is compact. In fact the ambient target need not be bounded: compactness of the boundary data confines the construction to a compact subtree.

## Definitions

Let  $Y$  be closed in a metric space  $X$  and let  $f : Y \rightarrow T$  be Lipschitz. A Lipschitz extension  $u : X \rightarrow T$  is an *absolutely minimal Lipschitz extension* (AMLE) if, for every open  $U \subseteq X \setminus Y$  and every Lipschitz  $h : X \rightarrow T$  agreeing with  $u$  on  $X \setminus U$ ,

$$\text{Lip}_U(h) \geq \text{Lip}_U(u).$$

For a map  $u : X \rightarrow T$ , say that  $u$  satisfies  *$T$ -comparison* on  $X \setminus Y$  when, for every  $t \in T$ , the real-valued function  $x \mapsto d_T(t, u(x))$  satisfies comparison with distance functions from above in the sense of [1, Section 2]. Proposition 7 of [1] shows that  $T$ -comparison implies the AMLE property. Importantly, the source notes that this implication works for an arbitrary metric target; the finite tree structure is needed only for the converse direction.

## Proof Intuition

The finite-tree theorem can be used as an approximation engine. The boundary image is compact, and in a complete  $\mathbb{R}$ -tree its closed convex hull is still compact. Approximate that hull from inside by finite subtrees and project the boundary values onto them. The source theorem produces AMLEs for all the finite problems. These maps are equi-Lipschitz and live in one compact tree, so a subsequence converges uniformly.

The key point is that the property used to recognize the limit is stable. If  $K$  is a finite approximating subtree and  $p(t)$  is the gate projection of  $t$  onto  $K$ , then for  $z \in K$ ,

$$d(t, z) = d(t, p(t)) + d(p(t), z).$$

Thus comparison for target points in  $K$  automatically gives comparison for every target point in the full compact hull. Uniform limits preserve these scalar comparison inequalities, and comparison implies absolute minimality.

## Compact Convex Hulls in $\mathbb{R}$ -Trees

**Lemma 1.** *Let  $T$  be a complete  $\mathbb{R}$ -tree and let  $A \subseteq T$  be compact. Then the closed convex hull*

$$C = \overline{\text{conv}}(A)$$

*is a compact  $\mathbb{R}$ -tree.*

*Proof.* Fix  $\varepsilon > 0$  and choose a finite  $\varepsilon$ -net  $F \subseteq A$ . In an  $\mathbb{R}$ -tree the convex hull of a set is the union of the geodesic segments joining pairs of its points. The set  $\text{conv}(F)$  is a finite union of compact segments and is therefore compact.

Let  $z \in \text{conv}(A)$ , so that  $z \in [a, b]$  for some  $a, b \in A$ . Choose  $a', b' \in F$  with  $d(a, a') < \varepsilon$  and  $d(b, b') < \varepsilon$ . Parameterize  $[a, b]$  and  $[a', b']$  proportionally and let  $z' \in [a', b']$  have the same parameter as  $z$ . Geodesics in an  $\mathbb{R}$ -tree are Busemann convex, hence

$$d(z, z') \leq (1-s)d(a, a') + sd(b, b') < \varepsilon$$

for the relevant  $s \in [0, 1]$ . Thus  $\text{conv}(A)$  lies in the  $\varepsilon$ -neighborhood of the compact set  $\text{conv}(F)$ . It follows that  $\text{conv}(A)$  is totally bounded. Its closure  $C$  is complete, being closed in the complete space  $T$ , and hence is compact. A closed convex subset of an  $\mathbb{R}$ -tree is again an  $\mathbb{R}$ -tree.  $\square$

## The Compact-Domain Extension Theorem

**Theorem 1.** *Let  $X$  be a compact length space, let  $\emptyset \neq Y \subseteq X$  be closed, let  $T$  be an arbitrary complete  $\mathbb{R}$ -tree, and let  $f : Y \rightarrow T$  be Lipschitz. Then  $f$  admits an AMLE  $u : X \rightarrow T$ . Moreover, writing*

$$C = \overline{\text{conv}}(f(Y)),$$

*one may choose  $u(X) \subseteq C$  and*

$$\text{Lip}_X(u) = \text{Lip}_Y(f).$$

*Proof.* Put  $L = \text{Lip}_Y(f)$ . Because  $Y$  is compact,  $f(Y)$  is compact; hence  $C$  is compact by Lemma 1.

Choose increasing finite sets  $F_n \subseteq C$  such that  $F_n$  is a  $1/n$ -net in  $C$ , and set

$$K_n = \text{conv}(F_n).$$

Each  $K_n$  is the realization of a finite graph-theoretical tree. Let  $p_n : C \rightarrow K_n$  be the nearest-point, or gate, projection. Closed convex subtrees of an  $\mathbb{R}$ -tree admit a 1-Lipschitz gate projection, and the net property gives

$$\sup_{z \in C} d(z, p_n(z)) \leq \frac{1}{n}. \quad (1)$$

Define  $f_n = p_n \circ f : Y \rightarrow K_n$ . Then  $\text{Lip}_Y(f_n) \leq L$ , and  $f_n \rightarrow f$  uniformly.

Theorem 1 of [1], applied to the compact length space  $X$  and finite metric tree  $K_n$ , supplies an AMLE  $u_n : X \rightarrow K_n$  of  $f_n$ . These maps may be taken with

$$\text{Lip}_X(u_n) = \text{Lip}_Y(f_n) \leq L. \quad (2)$$

For completeness, the upper bound also follows from absolute minimality: the injectivity of a metric tree gives an  $L$ -Lipschitz extension competitor; apply absolute minimality on  $X \setminus Y$ , and then use the length-space property and continuity to control pairs with one endpoint in  $Y$ .

We next regard every  $u_n$  as a map into the fixed compact tree  $C$ . We claim that  $u_n$  satisfies  $C$ -comparison, not merely  $K_n$ -comparison. Fix  $t \in C$  and put  $q = p_n(t)$ . For every  $z \in K_n$ , the gate property gives

$$d(t, z) = d(t, q) + d(q, z). \quad (3)$$

By Proposition 7 of [1], the scalar function  $x \mapsto d(q, u_n(x))$  satisfies comparison with distance functions from above. Adding the constant  $d(t, q)$  preserves comparison, so (3) shows that  $x \mapsto d(t, u_n(x))$  also satisfies comparison. Since  $t \in C$  was arbitrary, the claim follows.

The family  $(u_n)$  is equi-Lipschitz by (2), and it consists of maps from compact  $X$  into compact  $C$ . The Arzelà–Ascoli theorem [2, Theorem 6.1] therefore gives a uniformly convergent subsequence;

write  $u_n \rightarrow u : X \rightarrow C$ . Equation (1) implies  $u|_Y = f$ , and uniform convergence preserves the Lipschitz bound  $\text{Lip}_X(u) \leq L$ . The reverse inequality follows by restriction to  $Y$ .

We verify  $C$ -comparison. Fix  $t \in C$ . The real-valued functions

$$g_n(x) = d(t, u_n(x)) \quad \text{converge uniformly to} \quad g(x) = d(t, u(x)).$$

Comparison with distance functions from above is closed under uniform limits. Indeed, if  $g$  is bounded above by a distance cone on the boundary of an open set, then for every  $\delta > 0$  and all sufficiently large  $n$ ,  $g_n$  is bounded there by the cone plus  $\delta$ . Comparison for  $g_n$  propagates that bound to the open set; let  $n \rightarrow \infty$  and then  $\delta \downarrow 0$ . Hence  $u$  satisfies  $C$ -comparison. The comparison-implies-AMLE direction of Proposition 7 in [1] now shows that  $u$  is a  $C$ -valued AMLE of  $f$ .

Finally,  $C$  is a closed convex subtree of  $T$ , so its gate projection  $P_C : T \rightarrow C$  is 1-Lipschitz. If a  $T$ -valued competitor  $h$  agreed with  $u$  off an open set and had smaller Lipschitz constant there, then  $P_C \circ h$  would be a  $C$ -valued competitor with no larger Lipschitz constant. This contradicts absolute minimality in  $C$ . Thus  $u$  is also an AMLE with ambient target  $T$ .  $\square$

**Remark 1.** *If  $Y = \emptyset$ , any constant map is an AMLE after choosing a point of  $T$ . The nonempty assumption only avoids this harmless choice.*

## What Remains Open

Theorem 1 proves existence for the target-tree extension of the source conjecture when the domain is compact. It is stronger on the target side than the stated bounded- $\mathbb{R}$ -tree conjecture, because  $T$  itself may be unbounded. It does not prove:

- existence for arbitrary length spaces after local compactness is removed;
- existence for locally compact noncompact domains when the boundary range has noncompact convex hull in a bounded nonproper  $\mathbb{R}$ -tree;
- uniqueness of AMLEs for arbitrary  $\mathbb{R}$ -tree targets.

The last point is kept explicit because the source’s proof that AMLE implies tree comparison is written for finite metric trees. Our proof only uses the reverse implication, whose target generality the source explicitly records.

## Novelty Check and Human Review

The lightweight run indexes and bounded searches performed on 9 August 2026 used the arXiv identifier 1005.2535 and the exact paper title. Additional phrases included “R-tree AMLE,” “compact R-tree absolutely minimal Lipschitz extension,” and close variants. The searches found the source paper and general AMLE material, but no exact statement of Theorem 1. This is evidence, not a guarantee, of novelty.

Human review should focus on Lemma 1, the gate identity (3), stability of comparison under uniform convergence, and the ambient projection argument. The source-dependent point to verify is that comparison implies AMLE for an arbitrary metric target, as stated in the discussion preceding Proposition 7 of [1].

## References

- [1] A. Naor and S. Sheffield, *Absolutely minimal Lipschitz extension of tree-valued mappings*, Math. Ann. 354 (2012), 1049–1078; arXiv:1005.2535.
- [2] J. R. Munkres, *Topology: A First Course*, Prentice–Hall, 1975.